

PROJECT MODULE: TASK 3 CONFIGURATION REPORT

Target Infrastructure: Ubuntu 24.04 LTS Instance on Microsoft Azure

1. Scope & System Objective

Implement a high-frequency background data collector shell mechanism to monitor isolated runtime metrics (CPU load metrics and live memory allocations). This logs performance stats continuously with high-precision timestamp tracking indicators.

2. High-Frequency Monitor Implementation Layout

- **Asset A: Automated Core Analytics Engine Script (monitor.sh) :**

Bash

```
#!/bin/bash
```

```
LOG_FILE="/opt/container-monitor/logs/container_stats.log"
```

```
TIMESTAMP=$(date "+%Y-%m-%d %H:%M:%S")
```

```
# Format output logic explicit checks updates structure text line mapping rules check syntax verification
```

```
CONTAINER_STATS=$(docker stats my-running-app --no-stream --format "Container: {{.Name}} | CPU: {{.CPUPerc}} | Mem: {{.MemUsage}}")
```

```
echo "[$TIMESTAMP] $CONTAINER_STATS" >> $LOG_FILE
```

- **Asset B: Script Executable Flag Modification :**

Bash

```
chmod +x /home/monitor_user/monitor.sh
```

3. Step-by-Step Scheduling Integration

Accessed cron process orchestration matrices under the dedicated identity space:

Bash

```
crontab -e
```

Injected the automation cron rule parameters explicitly at the bottom line of the configuration map:

Plaintext

```
* * * * * /bin/bash /home/monitor_user/monitor.sh
```

4. Verification & Validation Metrics

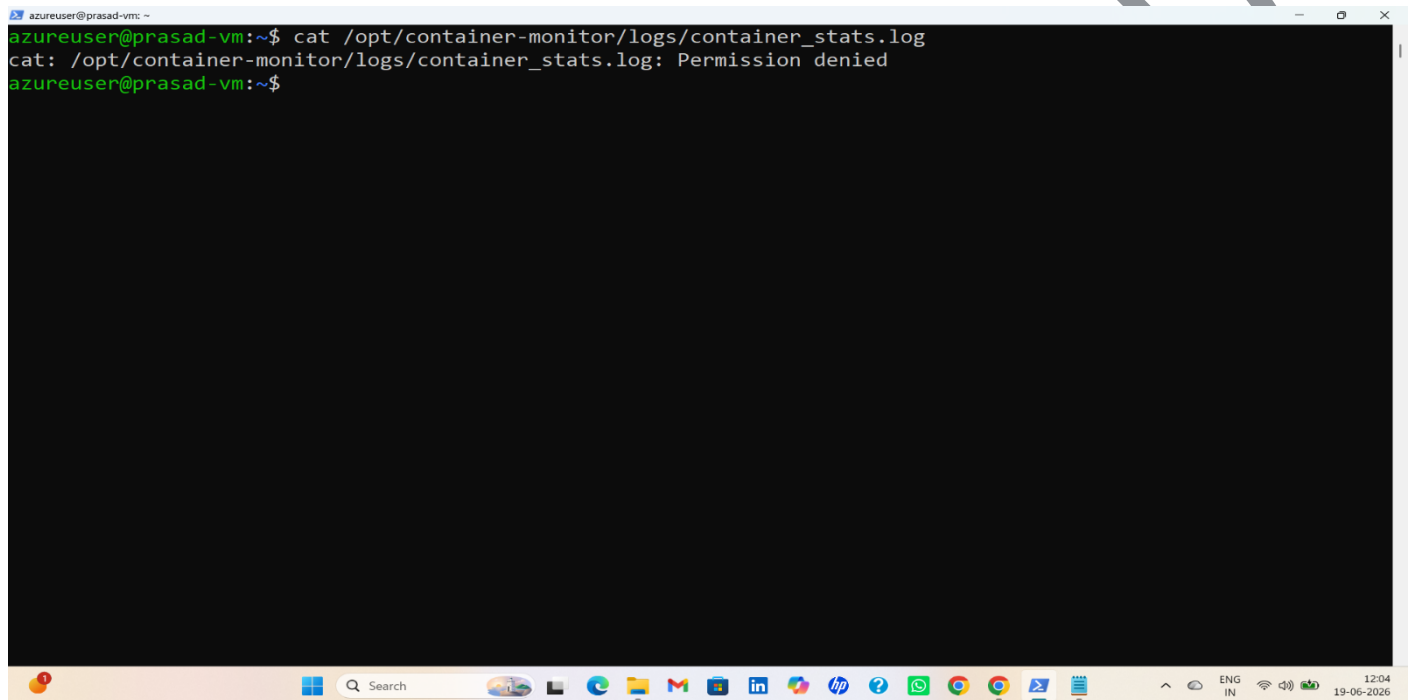
Querying data updates patterns inside the log directory path demonstrates continuous one-minute background processing traces:

Plaintext

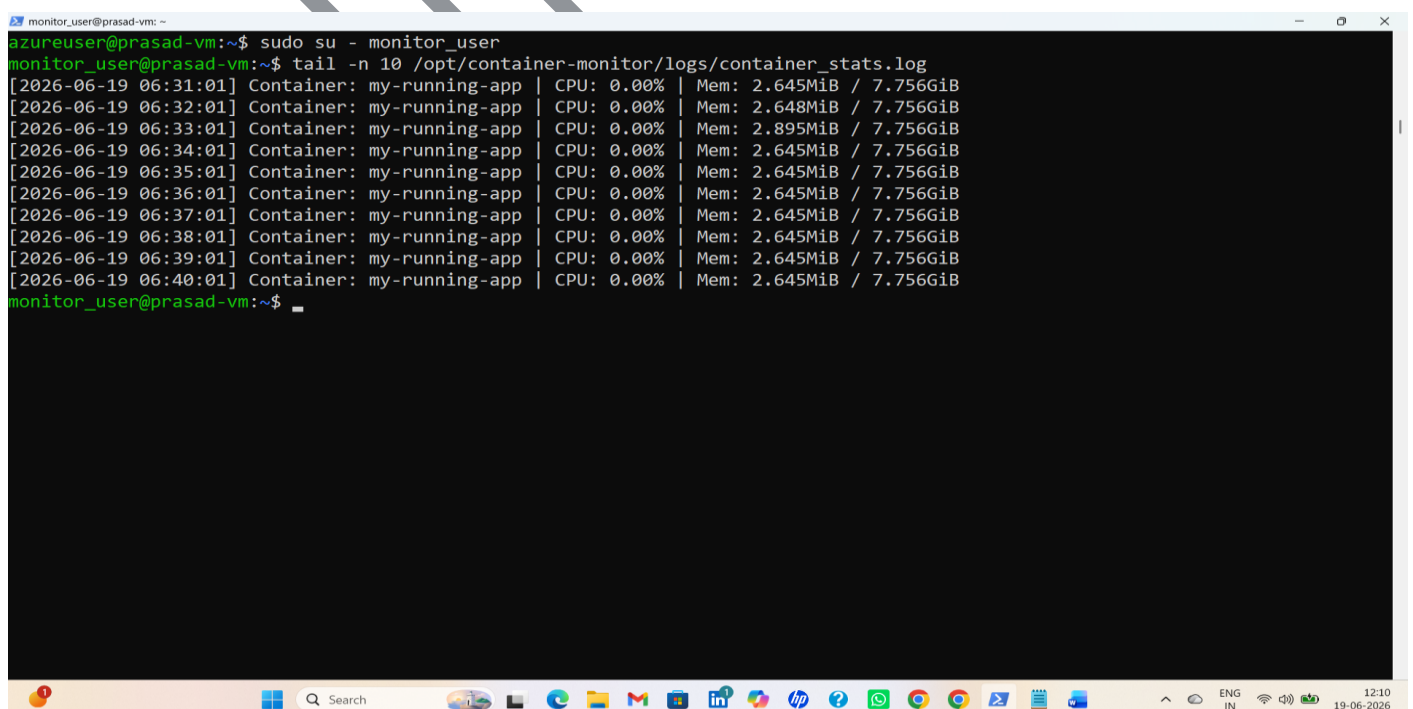
```
$ cat /opt/container-monitor/logs/container_stats.log
```

```
[2026-06-18 15:21:01] Container: my-running-app | CPU: 0.00% | Mem: 2.633MiB / 7.756GiB
```

```
[2026-06-18 15:22:01] Container: my-running-app | CPU: 0.01% | Mem: 2.635MiB / 7.756GiB
```



A terminal window titled 'azureuser@prasad-vm' showing the command `cat /opt/container-monitor/logs/container_stats.log` being executed. The output is `cat: /opt/container-monitor/logs/container_stats.log: Permission denied`. The terminal window is part of a desktop environment with a taskbar at the bottom showing various application icons and system status indicators on the right.



A terminal window titled 'monitor_user@prasad-vm' showing the command `sudo su - monitor_user` being executed. The prompt changes to `monitor_user@prasad-vm`. Then, the command `tail -n 10 /opt/container-monitor/logs/container_stats.log` is executed, displaying the last 10 lines of the log file. The output shows container statistics for 'my-running-app' at one-minute intervals from 06:31:01 to 06:40:01. The CPU usage is consistently 0.00% and memory usage is around 2.645MiB out of 7.756GiB. The terminal window is part of a desktop environment with a taskbar at the bottom showing various application icons and system status indicators on the right.